



Overview

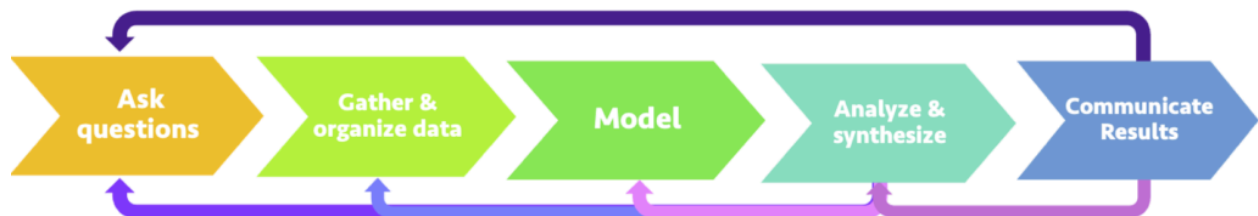
Data visualizations are critical to the data science process and provide a way to effectively communicate a data story. In this assignment, you will use the skills you learned from the course materials on data visualization to create meaningful data analysis on your survey data.

Assignment Preparation

Make a copy of this document to record your answers. Rename your copy of the document to **lastname_visualization**, where **lastname** is your last name. You will **place all answers and screenshots** in this Google Document. The steps below will tell you what to put in your answer document - use the spaces provided to insert your answers accordingly.

Part 1: Visualizing Your Survey Data

Good visualizations can help people make sense of data sets that are too large to interpret by looking at the raw data. In this part of the assignment, you will continue to analyze your survey data using Google Sheets to make visualizations of your own



Read the [Data Visualization 101 Guide](#)

- Familiarize yourself with the different chart types, paying particular attention to the scatter, line, and bar charts.



Complete the tutorial on [Data Visualizations](#)

- Provide the Share link to your completed tutorial (i.e your visualizations created from your copy of the “MovieRating_AvgRatingByAgeGender” data set).

Link:

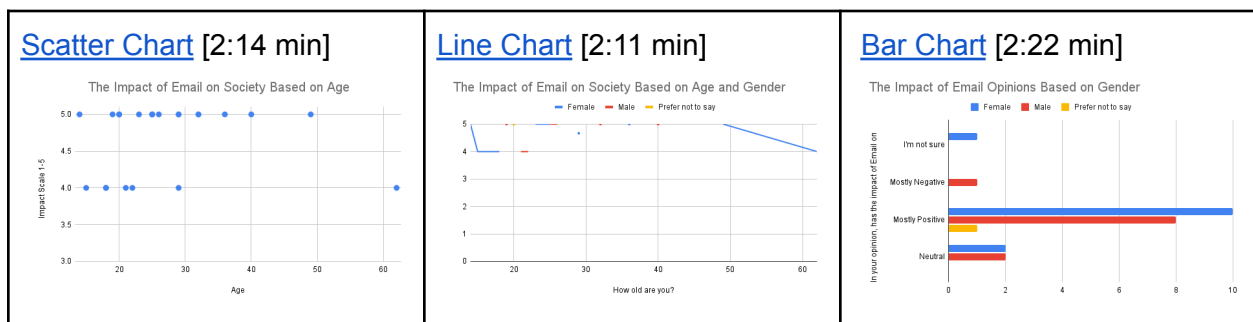
➤ Analyze and Synthesize ➤

You have learned to make scatter, line, and bar charts, but there are many other types of visualizations that people use to interpret data in different ways. You will now use these skills to create visualizations that further analyze the survey responses you collected in a previous assignment, using Google Sheets.

Note that this exercise is about using charts to explore trends in the data—not about creating the world's greatest chart!



Watch the video demonstrations on creating Data Visualizations using survey data:



- ➔ Open the working copy of your survey responses (from the previous assignment).
- ➔ Create a Scatter, Line, Bar, or other type of chart using your survey data.
- ➔ Your chart must include the following:
 - A title with your full name.
 - An appropriate legend.
 - Appropriate labels.



Provide a screenshot showing the chart you created from your survey data.

[Insert screenshot here]

→ Respond to the following question prompts:

What does your chart help you notice about the data?	
What connections and trends can you see in your chart?	

→ Search the Internet for a data visualization related to your final project topic.



Provide a screenshot of the visualization that you found.

[Insert screenshot here]

→ Respond to the following question prompts:

What search tool did you use to find the visualization? What search terms did you use?	
What is this visualization trying to communicate?	
Is this visualization effective? Why?	

Part 2: Summative Analysis

In this part of the assignment you will reflect on the entire data science process based on your final project topic including your original question, the data you collected from your survey, and the visualizations you created to draw conclusions from your data that support your analysis.

Consider the following:

- Revisit your original question: “What impact does [TOPIC] have on society?”
 - Will you change or refine your original question based on your findings?
 - What will your new question be?
- Revisit the automated visualizations/models of your survey data
 - What did these visuals help you notice about the data?
 - Do they provide evidence to support your question?
- Revisit the summary (pivot) table you created.
 - What connections and trends can you see in your summary table?
 - Does your summary table provide evidence to support your question?
- Revisit the original Scatter, Line, or Bar chart that you created.
 - What does the chart help you notice about the data?
 - Does it provide evidence to support your question?
- Answer the following question prompts to help analyze your data.

Summative Analysis	
What is your finalized Question?	
In 3-5 sentences, describe the process used to collect, model, and analyze your data (include the computing tools and techniques used in the process).	
In 3-5 sentences, describe a trend, pattern, or relationship	

that you found within your data (include a possible explanation for the trend or pattern).	
In 3-5 sentences, provide a summative conclusion based on your findings (i.e. what point about the impact of your computing innovation does your data analysis make?).	

Part 3: Communicating Your Data Story

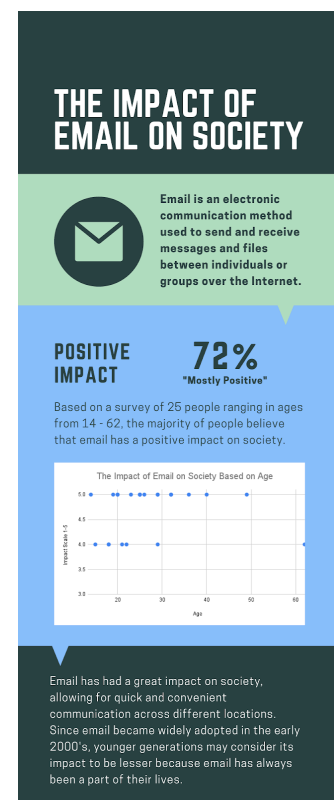
We collect data to learn about the world and test our hypotheses about the patterns and relationships that might otherwise be invisible. Computational tools make it possible to quickly collect and analyze vast amounts of data, but those processes still depend on knowledgeable and thoughtful people who understand the data and can clearly present the conclusions drawn from it.

➤ Communicate Results ➤

In this part of the assignment you will visualize your data story by creating an Infographic using Canva, a visual design tool, that clearly communicates the conclusions drawn from your survey data analysis (see [example](#)).

Consider the following:

- Keep it simple! Be brief and only include critical information that supports your data story.
- Make your infographic clear and easy to read.
- In some instances, a simple table may be a better way to show your data than a chart.
- Use text along with your visual to help explain your data story. Limit the amount of text you use - if you need to use too much text, ask yourself if there's a better way to visualize it.



- The infographic needs enough information/context to communicate a point to someone who is completely unfamiliar with this project

DO THE FOLLOWING:



Review the Canva tutorials:

- [Welcome Home](#) [1:34 min]
- [Start Designing](#) [1:40 min]
- [Easy Editing](#) [1:36 min]



Watch the [Infographic Demo using Canva](#) [6:19 min]

- Open Canva in a new browser tab
- Browse the Infographic template and select one to use as a starting point
- Begin editing the template using your own information based on your computing innovation topic
- Be sure to meet the **minimum requirements** as stated below:
 - Provide your question/impact statement
 - Include your name
 - Describe the computing innovation
 - Use at least one icon or graphic from canva
 - Reference your survey data (number of responses, age range, etc.)
 - Include at least one relevant statistic from your data
 - Include at least one original visualization from your analysis (chart, summary table, etc.)
 - Provide a summative conclusion that tells a story about your data



Provide a screenshot of your infographic.

[Insert screenshot here]



Provide a link to your infographic from Canva. (Click share at the top right then click “View-only link” and click copy.)

[Insert link here]